Case 49: WU-E uniform-community temporal convergence

1 Verification purpose

This case verifies time-step convergence of the intended Wildland–Urban ELMFIRE (WU–E) heat-accumulation and urban-propagation law with a fixed material fire-thermal property (FTP) on one fixed uniform-community grid. It refines the configured step through 4, 2, 1, and 0.5 s for families with low and high heat-release rate (HRR). The 30,000-kJ source branch is retained only as a plotted deviation diagnostic and is not an acceptance boundary.

Current result: FAIL. 8 of 8 required simulation conditions have evaluable evidence. Analysis procedure: **COMPLETE.**

2 Mathematical formulation and numerical implementation

The uniform-community and separation context follows the published WU–E parametric study [1]. The detailed intended equations are documented directly here. Executable semantics were checked against ELMFIRE spread-rate calculation and ELMFIRE level-set formulation at source revision `a2dfbcdf7220`. The reviewed components were identical to that revision; this identity claim does not extend to other modified model components. For received heat flux from direct flame contact (DFC) and radiation, $q = q_{\text{DFC}} + q_{\text{rad}}$, ELMFIRE forms

$$E = q\,\Delta t\,\Delta x^2. \tag{1}$$

The intended routine uses a fixed material property and

$$F_{\text{TP}} = F_{TP,crit} = 5400\ \text{kJ m}^{-2}, \qquad U = \frac{60E}{0.3048\,\Delta t\,\Delta x\,F_{TP,crit}} = \frac{196.85q\Delta x}{F_{TP,crit}}. \tag{2}$$

Thus Δt controls numerical integration but not the modeled material law. Heat is recomputed from every burning WU-E source before the two-stage level-set update. The pinned source's 30,000-kJ switch, dynamic FTP, and 35-mph divisor are documented deviations expected to degrade convergence.

3 Derivation of expected behavior

For a time-convergent observable $Q_{\Delta t}$,

$$Q_{\Delta t} = Q + C\Delta t^p + o(\Delta t^p), \qquad p \approx \log_2 \frac{\|Q_{2\Delta t} - Q_{\Delta t}\|}{\|Q_{\Delta t} - Q_{\Delta t/2}\|}. \tag{3}$$

The low-HRR family supplies a lower-amplitude heat-integration sequence. The high-HRR family supplies the time of arrival (TOA) observables used for propagation convergence. Both must have positive finite sampled heat; neither is classified by an energy branch in the intended model. ELMFIRE feedback level 1 records every solver-step start time. The evaluator requires the exact expected step count and increments within the log's 0.1 s print precision; any adaptive reduction or incomplete log makes the simulation condition NOT EVALUABLE.

On the fixed $\Delta x = 20$ m grid, the sampled heat-field diagnostic for every transient dump and physical fire behavior fuel model (FBFM) 91 cell is

$$E^*_{i,n} = (q_{\text{DFC},i,n} + q_{\text{rad},i,n})\Delta t_{\text{configured}}(20)^2.$$

The maximum sampled E_i^* is retained as a source-deviation diagnostic only. For the high-HRR landmark vector $\mathbf{t}$,

$$e_t = \frac{\|\mathbf{t}_1 - \mathbf{t}_{0.5}\|_2}{\max(\|\mathbf{t}_{0.5}\|_2, 10^{-12})},$$

with analogous scalar changes for fitted rate of spread (ROS) and total received heat. Observed order uses the 2, 1, and 0.5 s results.

4 Verification metrics and acceptance criteria

All eight exact-stop, input-matched simulation conditions are required, with generated namelists, tables, and rasters matching the case specification, complete transient stacks on the common 20 s dump schedule, and a complete logged solver-step sequence. Both HRR families must have positive finite sampled heat. Every expected logged start time from 0 through $t_{stop} - \Delta t$ must be present within 0.051 s, accommodating the log's 0.1 s print precision. For the two finest high-HRR simulation conditions, landmark-TOA and community-ROS changes must be at most 5%; total-heat change in each heat family must also be at most 5%. A resolvable nonzero sequence must have temporal order at least 0.5; a roundoff-flat three-level sequence is the explicit alternative, while a flat finest pair after a nonzero coarse change fails as possible stagnation. High-HRR TOA fits require $R^2 \geq 0.95$.

These are preregistered diagnostic criteria rather than literature-provided uncertainties. A 5-percent finest-pair limit is large enough to accommodate the half-second sampling/TOA quantization yet small enough to indicate practical time-step stability. The positive order floor of 0.5 tests contraction without claiming a formal order for this coupled clipped model, and $R^2 = 0.95$ rejects fragmentary propagation profiles before a single ROS is reported. The limits are not tuned after execution.

5 Simulation configuration and predicted outcome

The 240 m by 80 m community uses a fixed 20 m grid of fire behavior fuel model (FBFM) 91, a 40 m initially burning upwind strip, 20 m linear characteristic building dimension, and 10 m separation. The 15 mph wind is from the west. Building models 14 and 15 share a 30–270–300 s piecewise-linear design fire but use peaks of 20 and 800 kW m^{-2}, respectively. Runs end at 300 s. Firebrands, crown fire, stochastic wind, and suppression are disabled.

Quantity	Values	Units	Control intent
Timestep	4, 2, 1, and 0.5 s	s	Ratio-two temporal refinement
Grid / stop	20 / 300	m / s	Holds space and terminal time fixed
Wind / direction	15 / 270	mph / degree from	Fixed ellipse forcing
Low/high HRR peak	20 / 800	kW m^{-2}	Two amplitude convergence families
Terminal evidence	TOA and total heat	s / kJ	Chronology, ROS, and energy

Transient/log evidence	DFC, radiation, HRR; step timestamps	kW m^{-2} / s	Sampled heat and executed-step gates

Figure 1 is mechanically rendered from the prepared GeoTIFFs and the two fuel-table design fires. It precedes result plots and distinguishes prescribed inputs from calculated outputs.

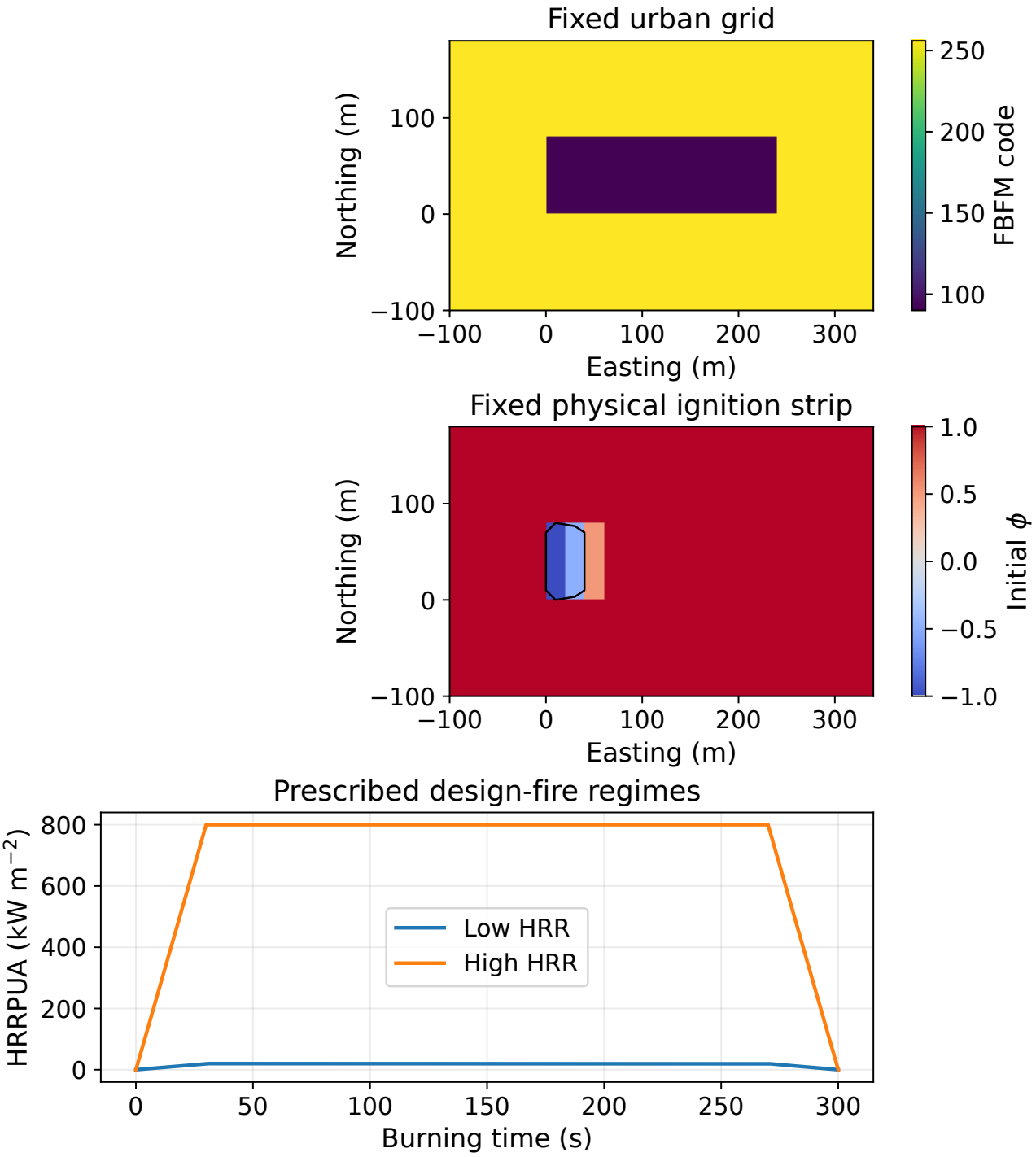


Figure 1: Prepared high-HRR FBFM and level-set rasters, plus the intended design-fire curves read from the case metadata. The postprocessor independently checks the copied building-fuel-model rows before any decision; these panels are inputs, not results.

6 Actual simulation results

The analysis procedure state is **COMPLETE**; the scientific decision is **FAIL**, with 8 of 8 required simulation conditions. Reason: One or more temporal-convergence acceptance components failed.. The executed binary SHA-256 is `ab47c5e338a3bc3b325df659b0764ce47d5b071e22d07817e4b5cab9f87da672`. For each completed simulation, the selected fields retain their actual output times and matching model, input, and simulation-record identities. All results must belong to those completed simulations.

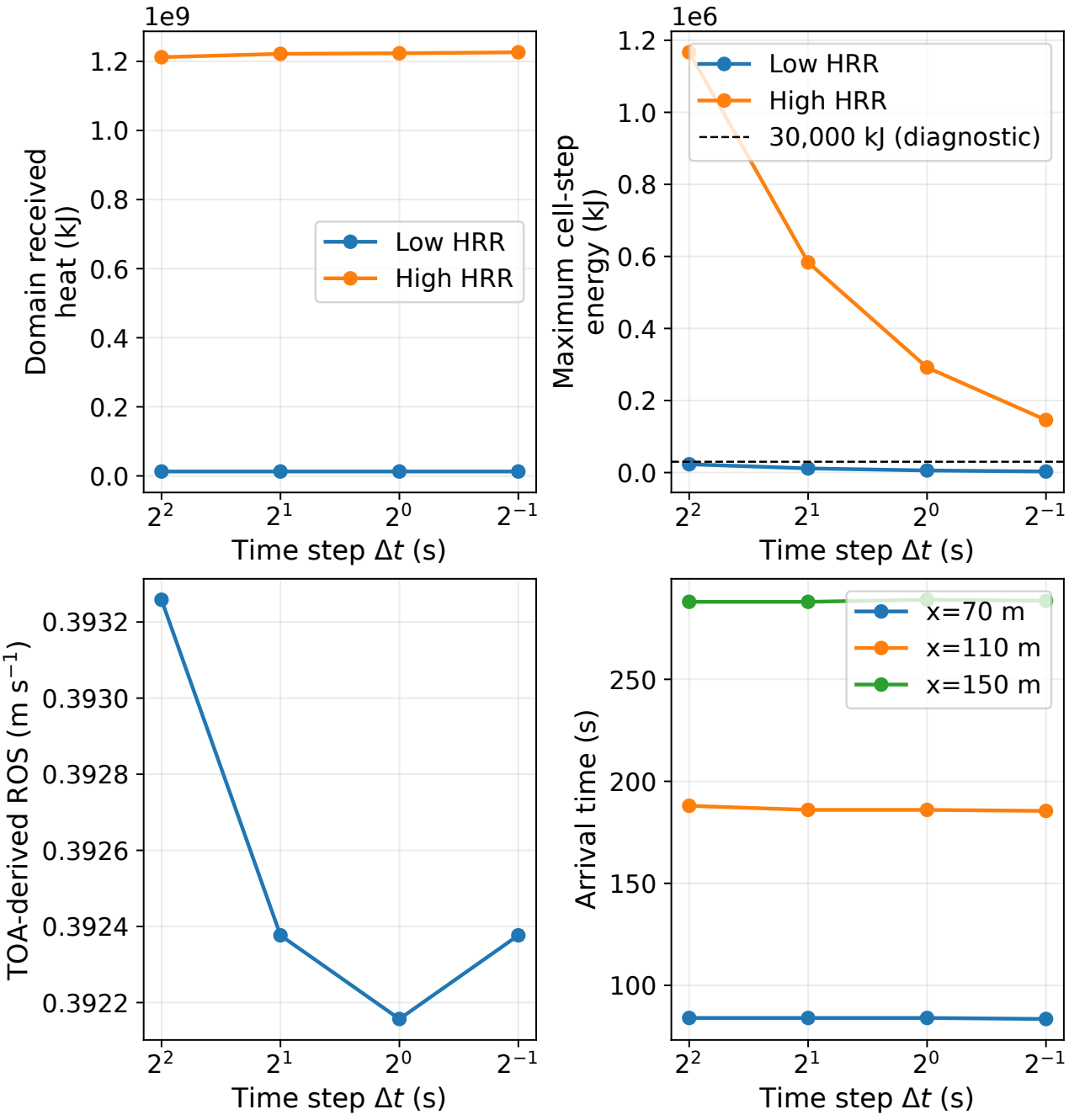


Figure 2: Exact common-time response of heat, sampled FBFM91 energy, ROS, and TOA from completed, input-matched simulations; convergence limits are 5%.

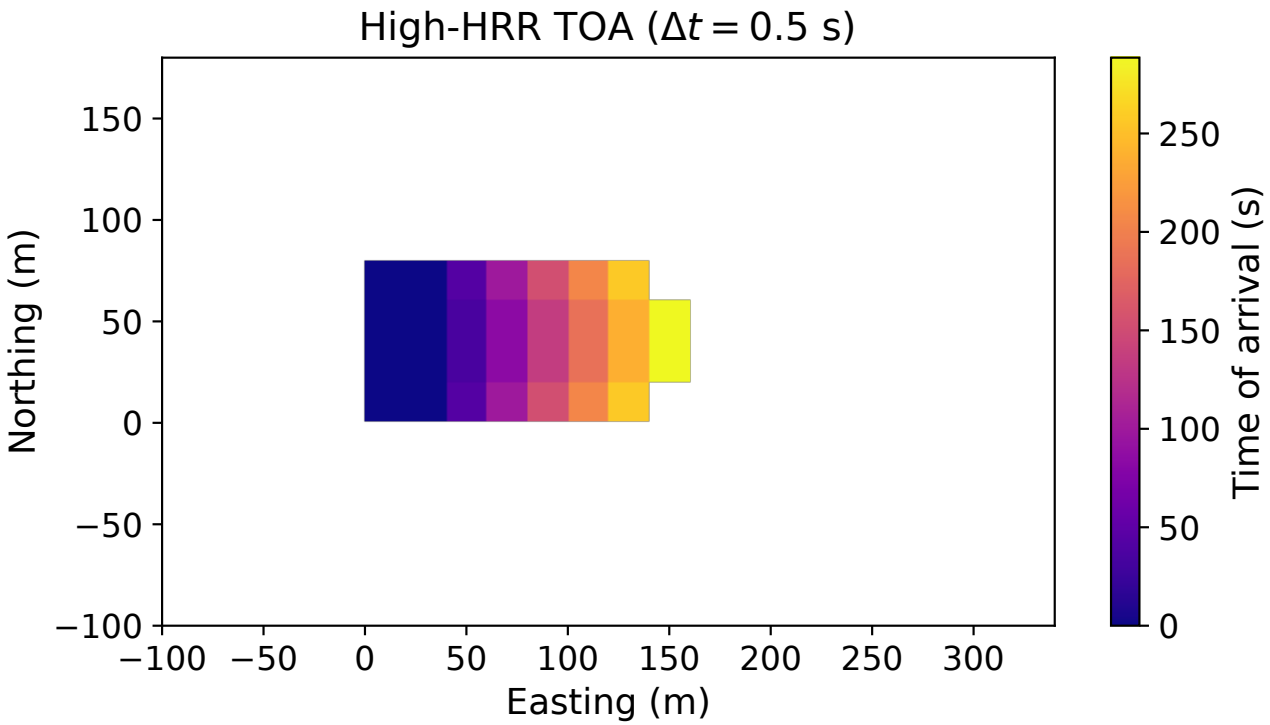


Figure 3: High-HRR exact-$t = 300$ s TOA for $\Delta t = 0.5$ s; the exact source raster is recorded in calculated verification and validation metrics.

7 Calculated metrics and verification decision

Current-input identity record output completeness — PASS

Criterion	8
Observed	*Units: simulation conditions.* 8

Low-HRR positive sampled FBFM91 cell-step energy — PASS

Criterion	> 0 and finite
Observed	*Units: kJ.* 23035.2

High-HRR positive sampled FBFM91 cell-step energy — PASS

Criterion	> 0 and finite
Observed	*Units: kJ.* 145850

Maximum logged solver-step deviation — PASS

Criterion	≤ 0.051
Observed	*Units: s.* 0

Finest-pair landmark TOA relative L2 change — PASS

Criterion	≤ 0.05
Observed	*Units: fraction.* 0.00245327

Finest-pair community ROS relative change — PASS

Criterion	≤ 0.05
Observed	*Units: fraction.* 0.000560224

High-HRR finest-pair total-heat relative change — PASS

Criterion	≤ 0.05
Observed	*Units: fraction.* 0.00237104

Low-HRR finest-pair total-heat relative change — PASS

Criterion	≤ 0.05
Observed	*Units: fraction.* 2.26103e-06

Landmark TOA observed temporal order — FAIL

Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	0.207519

Community ROS observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	0
High-HRR heat observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-0.883201
Low-HRR heat observed temporal order — FAIL	
Criterion	≥ 0.5 or roundoff-flat three-level sequence
Observed	-1.46388
Minimum high-HRR TOA-fit coefficient of determination — PASS	
Criterion	≥ 0.95
Observed	0.999798

The overall scientific decision is **FAIL**. Missing or stale products are NOT EVALUABLE; NOT RUN is retained only as a analysis procedure state and cannot pass. An incomplete propagation profile or a logged departure from the configured step produces NOT EVALUABLE rather than an invented convergence rate.

8 Technical interpretation

The complete step logs establish that the configured temporal sequence was executed. The finest high-HRR pair satisfies the relative-change limits, but the three-finest-level differences do not contract at the required rate. Practical closeness of the finest pair and a positive observed convergence order are different requirements; passing the first does not override failure of the second.

The source's step-energy switch and time-step-dependent replacement of FTP can change the modeled response as the step is refined. They are credible causes of poor convergence, but the finite sequence does not isolate a unique cause. The low-HRR heat changes are particularly small: its order is sensitive to float32 accumulation and to the evaluator's very tight flat-sequence test. That row should not independently be interpreted as a physical-model failure. The high-HRR order checks remain separate, decisive acceptance components.

9 Scope and limitations

This case verifies numerical behavior of the current executable against the intended fixed-FTP law, not empirical material calibration. The sampled cell-step energy is a source-deviation diagnostic only and does not determine acceptance. The case does not test explicit street gaps, heterogeneous structures, wildland transition rules, firebrands, or uncertainty in design-fire inputs.

References

[1] D. M. J. Purnomo et al., “Integrating an urban fire model into an operational wildland fire model to simulate one dimensional wildland–urban interface fires: a parametric study,” *International Journal of Wildland Fire*, 33 (2024), WF24102. doi:10.1071/WF24102.